

KENIL JITENDRABHAI KALATHIYA

M.Sc. Automotive Software Engineering

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PROFESSIONAL SUMMARY

Master's student in Automotive Software Engineering focusing on ADAS, simulation, driving systems, testing and functional safety (ISO 26262). Practical experience with **ROS2, Linux, Python, and distributed software systems**. Goal: Entry into automotive-related software development.

TECHNICAL SKILLS

Automotive & Embedded Software: ROS2, Linux, ADAS Basic, CAN & Ethernet

Model-Based Design & Simulation: MATLAB, Simulink

Functional Safety: ISO 26262 (Basic)

Computer Vision & AI: OpenCV, YOLOv8, NumPy, TensorFlow / PyTorch

Web Development: React.js, Node.js, Vue.js

Database: MongoDB, SQL

Version Control & Development Tools: GitHub, GitLab, Jira

Programming Languages: Python, C, C++, JavaScript, TypeScript

EDUCATION

Apr 2024 – Current	M.Sc. Automotive Software Engineering at Technische Universität Chemnitz , Germany NOTE: Grade 2.3
Jun 2019 – Jul 2023	B.Tech Information Technology at Gujarat Technological University , Ahmedabad, India NOTE: Grade 1.5

WORK EXPERIENCE

Apr 2026 – Present	Research Intern at Technische Universität Chemnitz - Designed a real-time structural health monitoring pipeline for UAV rotor arms (PX4 SITL, Gazebo Harmonic) - Fixed a sensor design issue by relocating strain gauges to arm root joints, decoupling bending and torsion signals - Built a 100 Hz ROS2 node logging bending, torsion and thrust data for ML training - Scripted MAVSDK flight maneuvers and a fault-injection dataset plan for CNN/LSTM classification Tech Stack: Python, PX4 SITL, Gazebo Harmonic, ROS2, MAVSDK, QGroundControl, Force-Torque Sensing, Machine Learning, UAV Simulation
Oct 2022 – Apr 2023	Software Developer (Intern) at Bigscal Technologies Pvt. Ltd. — Surat, India - Built frontend features and reusable UI components in React.js - Integrated the frontend with Flask-based REST APIs - Analyzed requirements, optimized features and ran unit/integration tests Tech Stack: React.js, JavaScript, Flask, REST APIs

PROJECTS

Sep 2026 – Present	Autonomous Drone Security Inspection System — YOLOv8 + ArduPilot SITL + Gazebo (<i>Ongoing</i>) • Built a simulated, unarmed inspect-and-report airport security drone (ArduPilot SITL, Gazebo Harmonic), inspired by the Aug 2026 Leipzig/Halle drone incidents • Ran live YOLOv8 detection on simulated camera feeds via gz-transport/OpenCV on a lightweight pymavlink/MAVSDK stack • Fixed VM/rendering issues (VirtualBox 3D acceleration, headless sensor rendering) for reliable GUI and headless operation Status: core pipeline functional; refining headless rendering for CI. Tech Stack: Python, YOLOv8, ArduPilot, Gazebo Harmonic, OpenCV, pymavlink/MAVSDK
Feb 2026 – Mar 2026	Intelligent Speed Adaptation (ISA) with Cruise Control — ADAS • Developed an ISA system with state-based cruise control and PID-based speed control • Implemented zone-based speed limits, validated under simulator and hardware constraints • Added RGB camera perception for real-time traffic sign recognition Tech Stack: Python, CARLA Simulator, OpenCV
Nov 2025	Vision-Based ADAS System — ADAS • Built lane detection and vehicle recognition with YOLO and OpenCV • Computed Time-To-Collision (TTC) and implemented forward collision warning • Extended the system with ROS2 nodes and basic camera–radar sensor fusion (simulation) Tech Stack: Python, OpenCV, YOLOv8, ROS2

STANDARDS AND PROCESSES

- ISO 26262 — Functional Safety (Basics)
- V-Model / Systems Engineering
- Software Testing & Quality Practices

CERTIFICATIONS

Microcredential: **Drone Systems Engineering for Safety-Critical Systems** — Dept. of Computer Engineering, TU Chemnitz [Aug – Sep 2026]

Coursera Certificate: **System Architecture** [2022]

SOFT SKILLS

- Team leadership and project management
- Independent work and quick adaptability
- Problem-solving mindset
- Strong communication and teamwork

LANGUAGES

English (Fluent – C1)
Deutsch (B1)